

Program	Diploma Engineering (DE)	Semester - 2
Type of Course	Professional Core	
Prerequisite	Basic Knowledge of C Programming.	
Course Objective	This hands-on C programming course provides advanced concepts of ANSI C language. Students are introduced to language elements like pointer, structure, union, user defined functions, pre-processor directives and file handling functions.	

Teaching Scheme (Contact Hours)				Examination Scheme				
Lecture	Tutorial	Lab	Credit	Theory Marks		Practical Marks		Total Marks
				SEE	CIA	SEE	CIA	
3	0	2	4	40	30	20	10	100

SEE - Semester End Examination, CIA - Continuous Internal Assessment (It consists of Assignments/Seminars/Presentations/MCQ Tests, etc.)

Course Content		T - Teaching Hours W - Weightage	
Sr.	Topics	T	W
1	Functions Introduction of user defined function (UDF), function declaration, definition, and call, categories of functions, call by value, call by reference, passing an arrays to functions, recursion, storage classes	9	20
2	Structure and Union Declaration and initialization of structure, accessing structure members, array of structure, arrays within a structure, nested structure, declaration and initialization of union, accessing union members, differentiate structure and union	8	20
3	Pointer and Dynamic Memory Allocation Introduction to pointers, initialization and declaration of a pointer, access a variable using a pointer, arithmetic operation on pointer, arrays of pointer, pointer to structure, introduction to dynamic memory allocation techniques, memory allocation and de-allocation functions	10	20
4	Pre-processor Directives Introduction to pre-processor directive, types of pre-processor directive, macros, simple macro, macro with arguments, file inclusion directives, compiler control directives, other directives (#undef and #pragma)	8	20
5	File Management Introduction to file management, file handling modes, file opening and closing functions, error handling, random access and input/output operation on files (getc(), putc(), getw(), putw(), fprintf(), fscanf()), command line argument	10	20
Total		45	100

Suggested Distribution Of Theory Marks Using Bloom's Taxonomy						
Level	Remembrance	Understanding	Application	Analyze	Evaluate	Create
Weightage	20	50	30	0	0	0

NOTE : This specification table shall be treated as a general guideline for the students and the teachers. The actual distribution of marks in the question paper may vary slightly from above table.

Course Outcomes

At the end of this course, students will be able to:

C01	perform C programs using function and storage class.
C02	implement structure and union programs using C programming language.
C03	solve problem using pointer and dynamic memory allocation in C programming language.
C04	implement macro and compiler control directive in C programming language.
C05	perform operation on a file using C programming language

Reference Books

1.	Programming in ANSI C By Balagurusamy,E Tata Mac-Graw Hill Education Private Limited
2.	Let Us C By Kanetkar,Yashavant P BPB Publications
3.	Mastering C By Venugopal,k r Tata Mac-Graw Hill Education Private Limited
4.	The C Programming Language By Kernighan Brain W; Ritchie Dennis M Pearson Education 2nd Edition

List of Practical

1.	Basic Programs Using User-Defined Function - Add two numbers using function. (A) - Find maximum and minimum between two numbers using function. (A) - Count simple interest using function. (A) - Generate Fibonacci series of N given number using function name fibbo(). (B)
2.	Intermediate Programs Using User-Defined Function - Return the maximum of three floating-point numbers. (A) - Return the minimum of three floating-point numbers. (A) - Swap two numbers using call by value and call by reference. (A) - Find all prime numbers between given interval using functions. (B) - Create a function that converts amount into words. (i.e. 9241: Nine Thousand Two Hundred Forty One) (C)
3.	Advanced Programs Using User Defined Function - Find factorial of a number using function and recursive function. (A) - Pass an array in function to print array elements. (A) - Find power of any number using recursion. (A) - Scan a character string passed as an argument and convert all lowercase string to uppercase string. (B) - Swap elements of two integer arrays using user define function. (B) - Find reverse of any number using recursion. (C)
4.	Basic Programs Using Structure and Union - Create, declare and initialize structure employee. (A) - Create a structure book with book title, author, publication, and price. Read data of 3 books and display. (A) - Demonstrate difference between structure and union. (A) - Demonstrate nested structure. (B) - Add two distances in feet and inches using structure. (C) - Add two times hh, mm and ss using structure. (C)
5.	Intermediate Programs Using Structure and Union Create structure student with name, percentage and age. Read data of 5 students using array of structure. (A)

	- Create structure student with name, percentage and age. Read data of N students using array of structure. Print details of student with maximum percentage. (A) - Create structure student with name, percentage and age. Read data of N students using array of structure. Arrange student data alphabetically and print all data. (B)
6.	Advanced Programs Using Structure and Union - Define a structure called Players which describe details like p_name, team, score, average. Write a program to read data for 3 players and print that data. (A) - Define a union called Players which describe details like p_name, team, score, average. Write a program to read data for 3 players and print that data. (A) - Define a structure called Players which describe details like p_name, team, score, average. Write a program to read data for N players and arrange p_name in alphabetical order. (B)
7.	Basic Programs Using Pointer - Print value and address of a variable. (A) - Demonstrate int, float, double and char pointer. (A) - Calculate sum of two numbers using pointer. (A) - Copy one array to another using pointers. (B) - Swap two arrays using pointers. (C)
8.	Advanced Programs Using Pointer - Swap value of two numbers using pointer. (A) - Store n elements in an array and print the elements using pointer. (A) - Add two matrix using pointers. (A) - Find length of string using pointers. (B) - Sort array using pointers. (C)
9.	Programs Using Dynamic Memory Allocation - Allocate and de-allocate memory for int, char and float variable at run time. (A) - Calculate the sum of n numbers entered by the user using malloc(). (A) - Calculate the sum of n numbers entered by the user using calloc(). (A) - Allocate dynamic memory for structure variable. (A)
10.	Programs Using Pre-processor Directives - Illustrate use of #define preprocessor (A) - Get current time using __TIME__ (A) - Define a function like Macro that should use printf Define Macro PRINT to print given integer argument. (A)
11.	Basic Programs Using File Handling - Create, open and close a file. (A) - Count chars, spaces, tabs and new lines in a file. (A) - Demonstrate functions fprintf(), fscanf(), fputc(), fgetc(), fseek() and rewind(). (A) - Print contents of file in reverse order. (B) - Capitalize first letter of each word in file. (B)
12.	Advanced Programs Using File Handling - Append one file at the end of other. (A) - Copy one file to another file. (A) - Merge alternate lines from two files. (B) - Delete all blank lines in a file then insert a blank line after the third line in a file.(C)

Miscellaneous

Useful Links

<https://www.geeksforgeeks.org/c-programming-language/>
<https://www.javatpoint.com/c-programming-language-tutorial>
<https://www.tutorialspoint.com/cprogramming/index.htm>